CHAPTER 10 / SUPPLEMENTARY MATERIAL

Cyber–Physical Security and Mission Assurance for Unmanned Aerial Vehicles

UAS System Deployment, Mission Assurance, and Information Dominance

Mission Planning and Operational Risk Assessment

Mission Planning and Operational Risk Assessment
Chapter 10 · UAS System Deployment, Mission Assurance, and Information Dominance

- **Plan the mission.** Define the objective, flight route, operating limits, and crew responsibilities.
- **Assess the hazards.** Consider airspace conflicts, ground exposure, weather, and loss of communications or navigation.
- **Verify mitigations.** Test contingency procedures, assess residual risk, and agree on launch and abort criteria.

Mini-exercise · Loss of command link
Use R=Likelihood x Severity on an illustrative 1–5 scale.
A verified contingency reduces the likelihood of leaving the planned flight area from 3 to 1.
Severity stays at 4.
Initial risk: 3x4=12 **Residual risk:** 1x4=4)

Threat-Informed Mission Planning Model

A UAV flies **Route A** east at (8 m/s) and a constant altitude of 40 m for 30s. It starts at (0,0,40) with heading measured from the positive (x)-axis.

1. Evaluate the mission state at (t=15 s) using Eq. (10.1).

$$\gamma_A(15) = [8(15),\, 0,\, 40,\, 8,\, 0^\circ] = [120\,\mathrm{m},\, 0\,\mathrm{m},\, 40\,\mathrm{m},\, 8\,\mathrm{m/s},\, 0^\circ].$$

2. Compare the inherent risk of command-link loss using Eq. (10.2).

On an illustrative 1–5 scale, both routes have likelihood (Li=3), vulnerability (Vi=2), and consequence (Ci=4). Route B spends less time in an area with poor link coverage, so its exposure score is lower.

Route	Exposure (Ei)	Inherent risk $(R_i^{\mathrm{inh}} = L_i V_i E_i C_i)$
A · Direct route	3	72
B · Alternative route	1	24

Spatial and Temporal Risk Along a Candidate Route

Spatial and Temporal Risk Along a Candidate Route
Worked exercise · Applying Eq. (10.3)
A UAV follows a 45-second route divided into three segments. Assume the inputs are constant within each segment. Use (Ac=20 {m^2}) and weights {wg,wa,ws,wc}=(2,3,1,2).

Segment	Duration Δt(s)	Ground density (ρg) (people/m^2)	Pair	Gsens	Pcyber
A · Open area	20	0.002	0.02	0.10	0.05
B · Populated area	10	0.010	0.06	0.20	0.10
C · Degraded link	15	0.004	0.03	0.15	0.20

Task. Calculate the cumulative route risk R_γ and identify the segment that contributes the most. Solution. Evaluate the integrand in each segment and multiply by the time spent there. For example, Segment A contributes

$$20[2(0.002)(20) + 3(0.02) + 0.10 + 2(0.05)] = 6.8 \quad .$$

$$R_\gamma = \underbrace{20(0.34)}_{\text{A}} + \underbrace{10(0.98)}_{\text{B}} + \underbrace{15(0.80)}_{\text{C}} = 6.8 + 9.8 + 12.0 = \boxed{28.6}.$$

Likelihood, Consequence, and Operational Risk Index

Worked exercise · Comparing two UAV routes
A UAV inspection team compares two routes for the hazard of **command-link loss**. Route B spends less time in poor link coverage. Both routes have the same likelihood, vulnerability, and consequence ratings for this exercise.

Route	Likelihood $\tilde{L}_i$	Vulnerability $\tilde{V}_i$	Exposure $\tilde{E}_i$	Consequence $\tilde{C}_i$
A · Direct route	0.60	0.40	0.60	0.80
B · Alternative route	0.60	0.40	0.20	0.80

Task. Use Eq. (10.4) with weights {wL,wV,wE,wC}=(0.25,0.20,0.30,0.25) to calculate each Operational Risk Index (ORI). Then classify each route using Eq. (10.5).
Worked solution. The weights sum to 1.00.

$$\mathrm{ORI}_A = 0.25(0.60) + 0.20(0.40) + 0.30(0.60) + 0.25(0.80) = \mathbf{0.61} \quad \text{High},$$
$$\mathrm{ORI}_B = 0.25(0.60) + 0.20(0.40) + 0.30(0.20) + 0.25(0.80) = \mathbf{0.49} \quad \text{Medium}.$$

Interpretation. Reduced exposure lowers the ORI by (0.12), moving Route B just below the **High** threshold of (0.50).

Detection, Delay, and Response Effectiveness

Worked exercise · Command-link loss during a UAV inspection

At a point along the route, three independent monitors have detection probabilities of (0.70), (0.60), and (0.50). The hazard has an inherent risk score of $R_i^{\text{inh}} = 72$ .Assume prevention effectiveness $M_i^{\text{prev}} = 0.25$, detection effectiveness $M_i^{\text{det}} = 0.50$ and secondary risk $R_i^{\text{sec}} = 3$

For this exercise, response effectiveness drops from (0.60) with a **2 s delay** to (0.20) with an **8 s delay**.

1. **Calculate the probability that at least one monitor detects the event using Eq. (10.6).**

$$P_{\text{det}} = 1 - (1 - 0.70)(1 - 0.60)(1 - 0.50) = \boxed{0.94}$$

2. **Compare residual risk for the two response delays using Eq. (10.7).**

Response delay	M_i^{resp}	Residual risk R_i^{res}
2 s	0.60	72(0.75)(0.50)(0.40)+3=13.8
8 s	0.20	72(0.75)(0.50)(0.80)+3=24.6

Interpretation. Both cases have a 94% detection probability. The longer delay raises residual risk by **10.8 points** because the response is less effective. Detection probability describes how often the system notices the event, while M_i^{det} represents the assumed risk reduction attributable to detection.

Operational Decision-Making

Worked exercise · Launch decision for a UAV inspection: The mission team uses illustrative thresholds of R_{acc}=10, R_{tol}=20, and R_{crit}=30. The **2 s** and **8 s** response cases use the residual-risk scores from the previous exercise. **Task.** Apply Eq. (10.8) to select the operational decision for each scenario.

Scenario	Residual risk R_i^{res}	Threshold check	Decision (D_i)
Baseline mission	10.0	≤10	Proceed
Response delay of 2 s	13.8	[13.8;20]	Proceed with controls
Response delay of 8 s	24.6	[20;30]	Modify or delay
Critical degradation	32.0	>30	Abort or activate response

Mission Assurance for UAV Operations

Exercise 1 · Mission assurance and residual risk
Eqs. (10.9)–(10.11)
During a UAV inspection, the navigation, command-link, and energy readiness scores are (0.92), (0.80), and (0.84), with weights {(0.40), (0.35), (0.25)}. The mission thresholds are θ_A=0.85 and θ_R=0.22.

Threat (b)	Posterior probability $P(b \mid z_{1:t})$	Impact I_b	Mitigation effectiveness η_b
Command-link degradation	0.60	0.50	0.60
Navigation degradation	0.40	0.40	0.50

Task. Calculate mission assurance A_m and residual mission risk R_m. Do both meet their thresholds?
Solution.

$$\mathcal{A}_m = 0.40(0.92) + 0.35(0.80) + 0.25(0.84) = \mathbf{0.858},$$
$$\mathcal{R}_m = 0.60(0.50)(1 - 0.60) + 0.40(0.40)(1 - 0.50) = \mathbf{0.20}.$$

Mission Assurance for UAV Operations

Exercise 2 · Flight constraints
Eq. (10.12)
The same UAV is **7 m** from the nearest obstacle and has **34 Wh** remaining. Its horizontal offset from the geofence center is (30,40) m. The minimum obstacle clearance is **8 m**, the required return-to-home energy is **30 Wh**, and the geofence radius is **60 m**.

Task. Check all three constraints and identify which one governs the decision.

Constraint	Check	Result
Obstacle clearance	(7<8 m)	Fail
Return-to-home energy	(34 >30 Wh	Pass
Geofence	$\sqrt{30^2 + 40^2} = 50 \leq 60\,\mathrm{m}$	Pass

Decision. Modify the route to restore obstacle clearance. The mission assurance and risk scores pass, but the current route violates the clearance constraint

Mission Assurance for UAV Operations

Exercise 3 · Swarm assurance under network degradation
Eq. (10.13)
Three UAVs have individual assurance scores of (0.858), (0.842), and (0.760). Let (α=0.75\). The network score Cnet drops from (0.90) to (0.70), while the individual scores stay fixed.

Task. Compute swarm assurance before and after the drop. Compare each result with an illustrative threshold of (0.80).
Solution. The mean individual assurance is (0.820).

$$\mathcal{A}_{\text{swarm,nominal}} = 0.75(0.820) + 0.25(0.90) = \mathbf{0.840},$$
$$\mathcal{A}_{\text{swarm,degraded}} = 0.75(0.820) + 0.25(0.70) = \mathbf{0.790}.$$

The network degradation alone moves swarm assurance below the threshold.

Information Dominance in Contested Environments

Exercise 1 · Information value and decision-cycle advantage
Eqs. (10.14)–(10.16)
A simulated UAV inspection team uses two information sources. In this exercise, (Q) is quality, (T) is timeliness, (A) is availability, and (D) is a degradation penalty. Use $\{\alpha, \beta, \gamma, \delta\} = (0.40, 0.30, 0.30, 0.20)$

Information source	Weight (w_k)	(Q_k)	(T_k)	(A_k)	(D_k)
Onboard telemetry	0.60	0.90	0.80	0.90	0.10
Ground observations	0.40	0.80	0.70	0.80	0.20

Task. Calculate the friendly information value, the Information Dominance Index (IDI), and the decision-cycle advantage. Use $V_I^{\text{adversary}} = 0.60$, $\epsilon = 0.001$ and friendly and adversary cycle times of (4 s} and (7 s) respectively.

Solution. The two source scores are (0.85) and (0.73).

Information Dominance in Contested Environments

$$V_I^{\text{friendly}} = 0.60(0.85) + 0.40(0.73) = \mathbf{0.802},$$
$$\text{IDI} = \frac{0.802}{0.802 + 0.60 + 0.001} \approx \mathbf{0.572},$$
$$\Delta T_{\text{adv}} = 7 - 4 = \mathbf{3}\,\text{s}.$$

The friendly team has the higher modeled information value and completes its decision cycle **3 s**.

Exercise 2 · Probability of success across two events, Eq. (10.17)

The team must detect, correctly classify, and mitigate two events during the mission.

Event	Detection	Classification given detection	Mitigation given correct classification
Command-link degradation	0.90	0.80	0.90
Navigation anomaly	0.80	0.75	0.90

Task. Assuming the two event outcomes are independent, calculate the probability that the team completes all three steps for **both** events.

$$P_{\text{ID}}(T) = (0.90 \times 0.80 \times 0.90)(0.80 \times 0.75 \times 0.90) = 0.648 \times 0.540 = \mathbf{0.34992} \approx \mathbf{35.0\%}$$

Information Dominance in Contested Environments

Exercise 3 · Weighted position fusion Eq. (10.18)
Three sources estimate the same UAV position at the same time in a common horizontal coordinate frame.

Source	Position estimate (x,y) in m	Fusion weight λ_k
UWB localization	(10,4)	0.50
Visual odometry	(12,6)	0.30
Inertial prediction	(14,5)	0.20

Task. Verify that the weights sum to one and calculate the fused position.

$$\hat{\mathbf{x}} = 0.50(10,4) + 0.30(12,6) + 0.20(14,5) = \boxed{(11.4,\ 4.8)\,\mathrm{m}}$$

Information Dominance in Contested Environments

Exercise 1 · Adaptive fusion weights — Eq. (10.19)
A UAV combines three position estimates. Let η ,(rk) be the source risk score, (Tk) timeliness, and (Ak) availability.

Source	(rk)	(Tk)	(Ak)
UWB localization	(0)	0.90	1.00
Visual odometry	(ln 2)	0.80	1.00
Inertial prediction	(ln 4)	0.80	0.50

Task. Calculate each fusion weight and verify that the weights sum to one.
Solution. The unnormalized weights $e^{-\eta r_k} T_k A_k$ are [0.90, 0.40,0.10]. Their sum is 1.40, so

$$(\lambda_1, \lambda_2, \lambda_3) = \frac{(0.90,\ 0.40,\ 0.10)}{1.40} \approx \boxed{(0.643,\ 0.286,\ 0.071)}.$$

Information Dominance in Contested Environments

Exercise 2 · Swarm information value — Eq. (10.20)
Three UAVs have individual information values $V_{I,i}=[0.80,0.70,0.60]$ and contribution weights $\rho_i=[0.5,0.3,0.2]$. The only nonzero cooperation terms are $c_{12}S_{12}=0.04$ and $c_{21}S_{21}=0.02$.
Task. Calculate the swarm information value and identify the contribution from cooperation.

$$V_{\text{swarm}} = \underbrace{0.50(0.80) + 0.30(0.70) + 0.20(0.60)}_{0.73} + \underbrace{0.04 + 0.02}_{0.06} = \boxed{0.79}.$$

The double sum uses **ordered pairs**, so it includes both 1→2 and 2→1.

Information Dominance in Contested Environments

Exercise 3 · Mission mode selection — Eq. (10.21)
Task. Use illustrative thresholds θ_1=0.65 and θ_2=0.45 to select the mission mode at each observation.

Observation	IDI(t)	Threshold check	Mission mode
Nominal information flow	0.72	(0.72 >0.65)	Normal mission execution
Reduced information quality	0.57	(0.45<0.57<0.65)	Degraded autonomous operation
Lower threshold reached	0.45	(0.45<0.65)	Degraded autonomous operation
Further degradation	0.40	(0.40<0.45)	Fail-safe or recovery mode

At exactly θ_2 the model still selects degraded autonomous operation. The transition to fail-safe or recovery occurs below that threshold.

Supply-Chain Security and Component Trustworthiness

Exercise 1 · Component inventory and attack surface
Eqs. (10.31)–(10.32)
A UAV inspection platform uses a flight controller, a command radio, and a payload computer. List the component set C and identify one example for each category in the supply-chain attack set A_{SC}.
Worked solution. The component set is *C={flight controller,command radio,payload computer}*

Category	Example
Hardware A_{HW}	Counterfeit flight-controller board
Firmware A_{FW}	Unauthorized radio firmware
Software A_{SW}	Compromised payload software dependency
Configuration A_{CFG}	Unauthorized change to a radio configuration
Update A_{UPD}	Tampered update package
Supplier A_{SUP}	Falsified supplier traceability records

Equation (10.32) takes the union of these categories. A single incident can appear in multiple categories, such as a tampered update that installs unauthorized firmware.

Supply-Chain Security and Component Trustworthiness

Exercise 2 · Component trust and supply-chain risk
Eqs. (10.33)–(10.34)

The team assigns the following illustrative trust scores on a 0–1 scale. Use weights $[w_p,w_f,w_s,w_b,w_v,w_u,w_a]=[2,2,1,1,1,2,1]$. In the risk calculation, Y_i is the component criticality weight, λ_i is the likelihood score, and I_i is the impact score.

Component	(P_i)	(F_i)	(S_i)	(B_i)	(V_i)	(U_i)	(A_i)	(Y_i)	(λ_i)	(I_i)
Flight controller	0.90	1.00	0.90	0.80	0.90	0.90	0.80	0.50	0.20	0.90
Command radio	0.80	0.70	0.70	0.60	0.70	0.70	0.60	0.30	0.40	0.80
Payload computer	0.70	0.60	0.60	0.50	0.60	0.60	0.50	0.20	0.30	0.60

Supply-Chain Security and Component Trustworthiness

Task. Calculate the trust score (Ti) for each component, then find the total supply-chain risk \(R_{\mathrm{SC}}\). Which component contributes the most risk?

Worked solution. The trust weights sum to 10. For the command radio, the weighted numerator is (7.00), so (Tradio)=7.00/10=0.70). Applying the same calculation to the other components gives (Tcontroller=0.90) and (Tpayload)=0.60\).

$$R_{\mathrm{SC}} = 0.50(0.20)(0.90)(1-0.90) + 0.30(0.40)(0.80)(1-0.70) + 0.20(0.30)(0.60)(1-0.60)$$
$$= 0.0090 + 0.0288 + 0.0144 = \boxed{0.0522}.$$

The **command radio** contributes the most risk, (0.0288), even though the payload computer has the lowest trust score. Equation (10.33) also accounts for criticality, likelihood, and impact.

Supply-Chain Security and Component Trustworthiness

Exercise 3 · Mission trustworthiness after a component improvement
Eq. (10.35)
Task. Calculate mission trustworthiness using the three component trust scores. Then recalculate it if the command radio's trust score improves from (0.70) to (0.85), with all other scores and criticality weights fixed.
Worked solution. The criticality weights sum to 1, so the normalized exponents are (0.50), (0.30), and (0.20).

$$T_{\text{mission,baseline}} = 0.90^{0.50}\,0.70^{0.30}\,0.60^{0.20} \approx \boxed{0.770},$$
$$T_{\text{mission,improved}} = 0.90^{0.50}\,0.85^{0.30}\,0.60^{0.20} \approx \boxed{0.816}.$$

Improving the radio's trust score raises the mission trust score by about **0.046**.

Risk Prioritization and Acceptance Criteria

Exercise 1 · Prioritizing component trust improvements
Eq. (10.36)
The UAV inspection platform from the previous exercise has three components. Assume each component's trust score (T_i) can be increased by (0.10) at the same cost, with all other inputs fixed.
Task. Calculate $\partial R_{SC}/\partial T_i = -\gamma_i \lambda_i I_i$ and identify which improvement gives the largest risk reduction.

Component	Criticality Y_i	Likelihood λ_i	Impact (I_i)	Risk sensitivity	Risk reduction for $\Delta T_i=0.10$
Flight controller	0.50	0.20	0.90	(-0.090)	0.0090
Command radio	0.30	0.40	0.80	(-0.096)	0.0096
Payload computer	0.20	0.30	0.60	(-0.036)	0.0036

Solution. Prioritize the **command radio**. For the same trust improvement, it yields the largest reduction in total supply-chain risk.

Risk Prioritization and Acceptance Criteria

Exercise 2 · Checking mission acceptance
Eq. (10.37)
The initial supply-chain risk is R_{SC}=0.0522). For this exercise, the mission risk limit $\tau_{mission}$ =0.0450. The flight controller and command radio are critical components, each with a minimum trust score of (0.85). The controller's trust stays at (0.90).
Task. Starting from a radio trust score of (0.70), check whether increasing it to (0.80) or (0.85) satisfies **both** acceptance criteria.
Solution. The radio's risk sensitivity is (-0.096), so the updated risk is R_{SC}=0.0522-0.096(T_{radio}-0.70).

Radio trust	Updated R_{SC}	Mission risk limit	Critical-component trust	Decision
0.70	0.0522	Fail	Fail	Reject
0.80	0.0426	Pass	Fail	Reject
0.85	0.0378	Pass	Pass	Accept

At (T_{radio})=0.80), the total risk is within the mission limit, but the radio still falls below its minimum trust score. At (0.85), both criteria pass.

Resilience, Redundancy, and Fault-Tolerant Design

Exercise 1 · Mission state and resilience
Eqs. (10.38)–(10.40)
During a UAV inspection, the primary command link degrades. At time (t_0), the mission team records each subsystem's operating state and assigns a resilience score $R_i(t_0)$ between 0 and 1.

Subsystem	Operating state ($x_i(t_0)$)	Weight (w_i)	Resilience ($R_i(t_0)$)
Flight control (f)	Nominal	0.30	0.95
Navigation (n)	Nominal	0.25	0.90
Communications (c)	Degraded	0.20	0.50
Payload (p)	Nominal	0.10	0.85
Energy (e)	Adequate reserve	0.15	0.80

Task. Write the mission state vector, verify the constraints in Eq. (10.40), and calculate ($R_m(t_0)$). Then recompute it after a backup link raises (Rc) from (0.50) to (0.80), with all other scores fixed.

Resilience, Redundancy, and Fault-Tolerant Design

Worked solution. The mission state vector is $x_m(t_0)$. The weights sum to 1, and all resilience scores lie in ([0,1]).

$$\mathcal{R}_{m,\text{primary}} = 0.30(0.95) + 0.25(0.90) + 0.20(0.50) + 0.10(0.85) + 0.15(0.80) = \mathbf{0.815},$$
$$\mathcal{R}_{m,\text{backup}} = 0.815 + 0.20(0.80 - 0.50) = \mathbf{0.875}.$$

The backup link raises mission resilience above the illustrative threshold $R_{min}=0.85$.

Exercise 2 · Probability of mission success
Eq. (10.41)
At a mission checkpoint, four mutually exclusive and exhaustive scenarios have the probabilities shown below. Use $R_{min}=0.85$, $d_{min}=8$ m, and $E_{RTH}=30$ Wh.
Task. Identify the scenarios that meet all three constraints and calculate ($P_{success}$). Recalculate it if the backup link is unavailable and only the communications-fault scenario changes.

Resilience, Redundancy, and Fault-Tolerant Design

Scenario	Probability	R_m	Clearance (d_{safe})	Remaining energy (E)	All criteria met?
Nominal operation	0.55	0.920	12 m	40 Wh	Yes
Communications fault, backup active	0.25	0.875	10 m	35 Wh	Yes
Reduced obstacle clearance	0.15	0.900	7 m	36 Wh	No
Low energy reserve	0.05	0.880	9 m	26 Wh	No

Worked solution. With the backup link, $P_{success}=0.55+0.25=0.80$. Without it, the communications-fault scenario falls to $R_m=0.815$, so only nominal operation passes and $P_{success}=0.55$. In this example, redundancy improves the modeled success probability by **25 percentage points**.

Reliability and Redundancy

Exercise 1 · Component and series reliability Eqs. (10.42)–(10.43)
A UAV inspection lasts **2 h**. Its command radio has a constant failure rate of $\lambda=0.02$ [1/h] . Over the same mission, the flight controller and navigation unit have reliabilities of (0.99) and (0.98). Assume independent failures and that all three components must work throughout the mission.
Task. Calculate the radio reliability and the reliability of the series system.

$$R_{\text{radio}}(2) = e^{-0.02(2)} \approx \mathbf{0.9608},$$
$$R_{\text{series}}(2) = 0.99(0.98)e^{-0.04} \approx \mathbf{0.9322}.$$

The probability that all three components complete the mission without failure is **93.22%**.

Reliability and Redundancy

Exercise 2 · Primary and backup radios Eq. (10.44)
The UAV now carries two independent radios, each with reliability (R=e^{-0.04}) over the 2 h mission. The command link remains available if at least one radio works. Assume fault detection and switching work whenever needed.
Task. Calculate the reliability of the redundant radio subsystem.

$$R_{\text{parallel}}(2) = 1 - (1 - e^{-0.04})^2 \approx \mathbf{0.9985}.$$

Under these assumptions, adding a backup raises radio subsystem reliability from **96.08% to 99.85%**.

Reliability and Redundancy

Exercise 3 · Triple modular redundancy Eq. (10.45)
Three independent processing modules each have mission reliability (R=0.90). An ideal majority voter produces the correct output when at least two modules work.

Task. Calculate the reliability of the triple modular redundancy (TMR) system

$$R_{\mathrm{TMR}} = 3(0.90)^2 - 2(0.90)^3 = \mathbf{0.972}.$$

The majority-voting system has a reliability of **97.2%**.

Reliability and Redundancy

Exercise 4 · Availability and repair time Eq. (10.46)
A repairable command radio has a mean time to failure (MTTF) of 50 h. A maintenance change reduces the mean time to repair (MTTR) from 2 h to 0.5 h.
Task. Calculate the long-run availability before and after the maintenance change.

Maintenance case	Availability (A)
Before	(50/(50+2)=96.15%
After	(50/(50+0.5)=99.01 %

Reducing repair time increases availability by **2.86 percentage points**. Availability measures the long-run fraction of time the radio is operational. Mission reliability measures the probability that it completes a specified mission without failure.

Post-Mission Analysis and Security Auditing

Exercise 1 · Audit evidence inventory Eq. (10.61)
After a UAV inspection, the team collects flight-control logs (L_f), command-link logs (L_c), RF-monitoring logs (L_{rf}), sensor logs (L_s), operator logs (L_o), and mission-management logs (L_m). The payload log (L_p) is missing.
Task. Compare the collected records with the expected audit set A_{pm}, identify the evidence gap.

$$\mathcal{A}_{\text{pm}} = \{L_f, L_c, L_{rf}, L_s, L_p, L_o, L_m\}, \qquad \mathcal{A}_{\text{pm}} \setminus \mathcal{A}_{\text{collected}} = \boxed{\{L_p\}}.$$

Result. Six of the seven expected log types are available. The missing payload log limits the team's ability to verify when the payload stopped recording.

Post-Mission Analysis and Security Auditing

Exercise 2 · Timeline reconstruction and event correlation Eqs. (10.62)–(10.63)
Three systems record events on different clocks. A positive offset Δ_k means the source clock is ahead of the reference clock. Use a correlation window of τ_c=0.10.

Task. Correct each timestamp using $\hat{t}_i^{(k)} = t_i^{(k)} - \Delta_k$. Which events fall within the correlation window of the command-link warning?

Event	Recorded time $t_i^{(k)}$	Clock offset Δk	Corrected time $\hat{t}_i^{(k)}$
Command-link warning	120.46 s	(+0.18) s	120.28 s
RF power increase	120.14 s	(-0.10) s	120.24 s
Operator acknowledgment	120.51 s	(+0.05) s	120.46 s

$$|\hat{t}_{\text{warning}} - \hat{t}_{\text{RF}}| = |120.28 - 120.24| = 0.04\,\text{s} \le \tau_c,$$
$$|\hat{t}_{\text{warning}} - \hat{t}_{\text{ack}}| = |120.28 - 120.46| = 0.18\,\text{s} > \tau_c.$$

Result. Only the RF power increase falls within the window. Its timing supports further investigation into the link warning's cause.

Post-Mission Analysis and Security Auditing

Exercise 3 · Detecting an altered audit record Eq. (10.64)
The flight controller stores three records r_1,r_2,r_3 in a hash chain. The auditor has a known starting value (h_0) and a trusted copy of the final hash (h_3). During review, record (r_2) is found to have been replaced by (r'_2).

Task. Write the verification equations and explain how the auditor detects the alteration.

$$
\begin{aligned}
h_1 &= H(r_1 \parallel h_0 \parallel t_1 \parallel ID_{\text{sys}}),\\
h'_2 &= H(r'_2 \parallel h_1 \parallel t_2 \parallel ID_{\text{sys}}),\\
h'_3 &= H(r_3 \parallel h'_2 \parallel t_3 \parallel ID_{\text{sys}}).
\end{aligned}
$$

Result. Changing (r_2) changes the input to the second hash and propagates through the third hash, even though (r_3) is unchanged. A mismatch $h'_3 \neq h_3$ flags an integrity failure. The final hash comparison identifies a mismatch in the chain. Locating the first altered record requires a trusted intermediate hash or an independent copy of the records.

Anomaly Scoring for Drone Attack Detection

Exercise 1 · Anomaly alert and attack probability
Eqs. (10.67)–(10.68)
During a simulated UAV inspection, the anomaly detector reports St=3.20 with a threshold Y=3. A separate classifier uses standardized RF power, packet loss, and navigation residual features z=[2.0,1.0,1.0], weights w=[0.8,0.6,0.4], and bias b=-1.5.

Task. Determine whether the detector raises an alert and calculate the classifier's estimated attack probability.

Worked solution. The detector raises an alert because (3.20>3.00\). With $\sigma(u) = 1/(1+e^{-u})$

$$P_{\text{attack}} = \sigma(0.8(2.0) + 0.6(1.0) + 0.4(1.0) - 1.5) = \sigma(1.10) \approx \boxed{0.750}.$$

The alert flags unusual behavior. The estimated probability can help prioritize review, while incident confirmation requires corroborating evidence.

Anomaly Scoring for Drone Attack Detection

Exercise 2 · Mission security los Eq. (10.69)
A simulated mission lasts (200 s), with a =10 s response delay. The UAV deviates 20 m over a planned route length of 200 m. Use C_{data}=0.90, A_{link}=0.80, R_{safety}=0.25, and weights α=[0.20,0.25,0.20,0.15,0.20].
Task. Calculate the security loss and identify its largest weighted contribution.

$$\begin{aligned}\mathcal{L}_{\text{sec}} &= 0.20\frac{10}{200} + 0.25\frac{20}{200} + 0.20(1-0.90) + 0.15(1-0.80) + 0.20(0.25) \\ &= 0.010 + 0.025 + 0.020 + 0.030 + 0.050 = \boxed{0.135}.\end{aligned}$$

The **safety risk term** contributes the most, at 0.050.

Anomaly Scoring for Drone Attack Detection

Exercise 3 · Evidence completeness and forensic confidence Eqs. (10.70)–(10.71)
The auditor expects three records. For this exercise, a missing record receives a quality score of zero.

Evidence	Available?	Quality (q_i)	Confidence weight (β_i)	Completeness weight (w_i)
Flight-control log	Yes	0.90	0.50	5
Command-link log	Yes	0.80	0.30	3
RF-monitoring log	No	0.00	0.20	2

Task. Calculate forensic confidence (C_F) and weighted evidence completeness η_E

$$C_F = 0.50(0.90) + 0.30(0.80) + 0.20(0) = \boxed{0.69},$$
$$\eta_E = \frac{1(5) + 1(3) + 0(2)}{5 + 3 + 2} = \boxed{0.80}.$$

Completeness measures how much expected evidence was collected. Forensic confidence also accounts for the quality of that evidence.

Anomaly Scoring for Drone Attack Detection

Exercise 4 · Residual risk after mitigation Eq. (10.72)
The initial risk of command-link loss is $R_{initial}$=0.30. In the simulation, a backup link provides mitigation effectiveness E_{mit}=0.60.

Task. Calculate residual risk and compare it with an illustrative acceptance limit of (0.15).

$$R_{\text{residual}} = R_{\text{initial}}(1 - E_{\text{mit}}) = 0.30(1 - 0.60) = \boxed{0.12}.$$

The residual risk meets the exercise's acceptance criterion because (0.12<0.15).